Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/03/2017
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Colin, Catherine Ph.D. (Institut de Mécanique des Fluides de Toulouse)		
Grant/Contract No.:	NNX15AK16G		
Performance Goal No.:			
Performance Goal Text:			

EDITOR'S NOTE (5/17/2016): Continuation of "Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry" with same Principal Investigator, Dr. Jungho Kim, grant # NNX12AF31A. Added to Task Book when received information.
This effort will be conducted with the European Space Agency (ESA) in one of the facilities/test articles that they are developing.

Fluorescent paints that are sensitive to oxygen concentration (pressure sensitive paints, PSP) and temperature (TSP) have been used to measure shear stress, surface pressure, and wall temperature distributions in aerodynamic and fluid mechanics applications since the 1980s. It has seen very limited use for phase-change applications, however. If only temperature is to be measured, the paint is encapsulated within a binder or between impermeable barriers to eliminate sensitivity to oxygen. The molecules are typically stimulated using blue-UV light, and emission at a longer wavelength

Task Description:	occurs. The intensity of the emitted light decreases with temperature and oxygen concentration, and a filter is used to remove the excitation light before the emitted light is measured by a camera. The emission lifetime is typically on the order of microseconds, much faster than the boiling processes being measured, so the system frequency response is usually limited by the thermal response of the materials onto which the paint is bonded. The advantages of TSPs over infrared thermography and other techniques such as microheater arrays or microbolometers to measure temperature are numerous. Readily available CCD or CMOS cameras can be used to measure the intensity distribution instead of expensive IR cameras. Since TSPs respond very quickly, the transient response of TSP based measurements is limited in practice only by the speed of the camera, which are typically much faster than IR cameras. Also, since visible light is used for illumination and sensing, glass, clear plastics, PDMS and other commonly used materials can be used in place of expensive and difficult to machine IR transparent materials (e.g., silicon, sapphire, CaF2) as substrates. Calibration of TSPs can be performed without having to consider emission from the transparent substrate they are placed on, unlike for IR measurements. Finally, the wavelength of the light used (~0.5 mm for TSPs) is much smaller than the 3-5 µm or 8-12 µm light IR cameras are sensitive to, allowing much higher spatial resolution measurements. The diffraction limit for 0.6 µm light in air is ~0.3 µm. TSPs have quite high sensitivity (approximately -2%/°C), and TSPs are available that can be used over a wide temperature range. Although TSPs have been used for aerospace applications for many years, very little work has been performed using this technique for two-phase flows. We have developed methods whereby temperature and heat flux distributions can be measured during pool boiling and flow boiling so models of pool and flow boiling can be verified. A test apparatus has been developed and this technique will be used to measure gravity effects on flow boiling during a series of low-g aircraft flights in March, 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	A TSP based pool boiling test apparatus was developed to measure the temperature variation during pool boiling of HFE7100 at 1 atm. Germanium dots were deposited using a shadow mask in onto one side of an adhesive to serve as an opaque layer. TSP was brushed onto the germanium dots, then the adhesive was attached to the sapphire substrate so the temperature distribution at this surface could be measured. Another TSP layer was then deposited onto one side of an adhesive, then attached to the first adhesive to encapsulate the TSP. To form an opaque heating layer, a NiCr film with an electrical resistance of ~25 O/square was attached. A DC power supply was used to heat the NiCr heater, and power was measured using a four-wire technique. A 12-bit, high-speed CMOS camera (Phantom Miro eX4, 800x600 pixels) was used to monitor the TSP intensity. A long pass filter (Midopt LP590-58) in front of the camera was used to block any extraneous light from the UV LED (Cree XTE 400 nm ultraviolet). A photomultiplier was used to correct for any variations in the LED output once the TSP was calibrated. The test section was placed within a black enclosure to minimize external light influences. Photobleaching is a phenomena where the TSP is permanently damaged by the UV light, resulting in a decrease in emission intensity with time. This was found to be fairly serious in our test apparatus. For example, at the UV LED intensity for which data was obtained, 1 hour of continuous exposure resulted in a 6% decrease in TSP emission intensity. Calibration of the TSP was performed by circulating 3M HFE 7500 (Tsat=128°C at 1 atm) within the test chamber between 40-100°C and measuring the TSP intensity using the camera at varying LED brightness to bracket the expected LED intensity during the tests. The intensity distribution was averaged over 100 frames to minimize camera noise. A 5th order polynomial was fit to this data and was accurate to 0.5°C. Calibration curves indicated the emission intensity decreased with temperature at a rate of about -3%/°C. Data obtained over a range of heat fluxes through CHF indicated the viability of TSP based measurements. We are implementing TSP use in the flow boiling apparatus and expect to obtain data in variable gravity environments during the March, 2017 parabolic flights.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Articles in Peer-reviewed Journals	Moaveni S, Kim J. "An inverse solution for reconstruction of the heat transfer coefficient from the knowledge of two temperature values in a solid substrate." Inverse Problems in Science and Engineering. 2017;25:129-53. http://dx.doi.org/10.1080/17415977.2016.1161035 , Jan-2017